Hyperparameter	Values
Layers	{1, 2, 4}
Nodes per layer	{64, 128, 256, 512}
Dropout	[0, 0.7]
Weigh decay	{0.4, 0.2, 0.1, 0.05, 0.02, 0.01, 0}
Batch size	{64, 128, 256, 512, 1024}
α (DeepHit)	[0, 1]
σ (DeepHit)	{0.1, 0.25, 0.5, 1, 2.5, 5, 10, 100}
Num. durations (DeepHit)	{50, 100, 200, 400}
λ (Cox-Time and Cox-MLP (CC))	{0.1, 0.01, 0.001, 0}
Log durations (Cox-Time)	{True, False}
Ridge (Cox (Linear))	{1000, 100, 10, 1, 0.1, 0.01, 10^{-3}, 10^{-4}, 10^{-5}}
Split covariates (RSF)	{2, 4, 6, 8}
Size leaf (RSF)	{2, 8, 32, 64, 128}

Table A.1: Hyperparameter search space for experiments on Rot. & GBSG, SUPPORT, METABRIC, and FLCHAIN.

Hyperparameter	Values
Layers	{4, 6, 8}
Nodes per layer	{128, 256, 512}
Dropout	{0, 0.1, 0.5}
α(*)	{0, 0.001, 0.1, 0.2, 0.5, 0.8, 0.9, 0.99, 0.999, 1}
σ(*)	{0.01, 0.1, 0.25, 0.5, 1, 10, 100}
Log durations(*)	{True, False}
Split covariates	{2, 4, 6, 8}
Size leaf	{8, 10, 20, 50}

Table A.2: KKBox hyperparameter configurations. (*) denotes parameters found with a two layer network with 128 nodes.

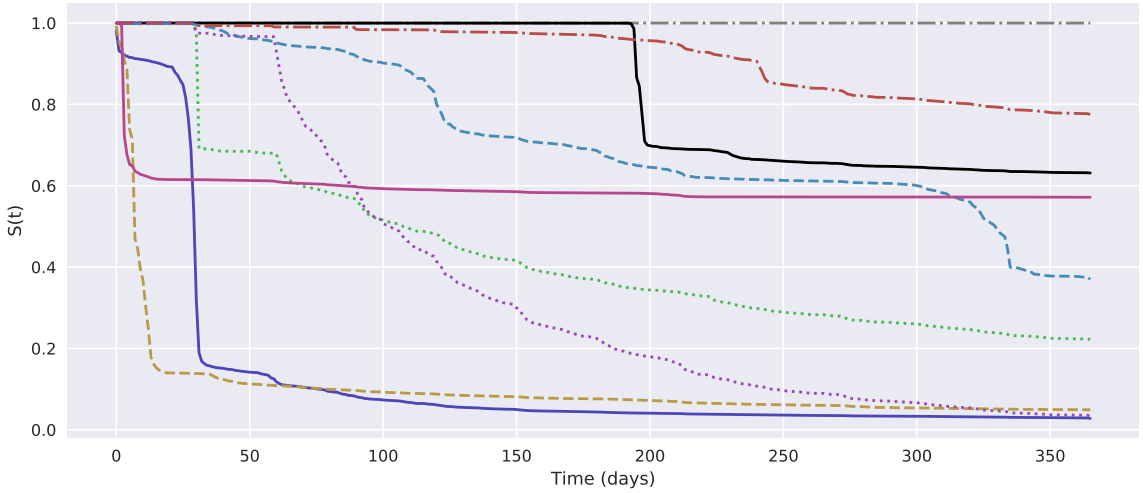


Figure B.1: Survival curves from the case study in Section 6.2, which models the times at which customers of a streaming service churn. Each curve gives the estimated probabilities for a customer not having churned (probabilities of still being a customer at any given time). The time axis shows the number of days since first subscription. The curves are generated by the Cox-Time method from Section 3.4.

loss function of DeepHit, and we assumed it should generalize well across network structures. The same goes for whether or not we should log-transform the durations passed to Cox-Time. Hence, to reduce the hyperparameter search, we found suitable values with a smaller network.

For the proposed Cox-MLP (CC) and Cox-Time, we used a fixed penalty $\lambda = 0.001$ in (10). All networks were trained with batch size of 1028, and the best performing architectures can be found in Table 6.

Appendix B. Details from KKBox Churn Case Study

In the following, we provide some details of the KKBox case study that were exempt from the main article.

In Figure B.1, we show an example of nine survival curves estimated by Cox-Time on the KKBox data set. Each line represents an individual from the test set. It is clear that the Cox-Time method has learned to represent a variety of survival curves.

B.1 KKBox Data Set

KKBox provides data consisting of general customer information (city, age, gender, initial registration), transaction logs listing how customers manage their subscription plans, and user logs containing some aggregate information of the customers' usage of the streaming

service. KKBox defines a customer as churned if he or she has failed to resubscribe to their service more than 30 days after their last subscription expired.

Through the transaction information, we can create a data set with survival times and censoring indicators. We keep KKBox's original churn definition of 30 days absence and calculate the survival time from the date of the first subscription or the earliest record of the customer in the transaction logs. Customers that have previously churned but resubscribed, are treated as new customers with some covariate information describing their previous subscription history.

All covariates are calculated at the time of subscription of each customer, and they do not change with time. This is a simplification, as the covariates of a customer are typically not stationary. However, as our objective is to evaluate the proposed methodology, we refrain from doing extensive feature engineering. In this regard, we further disregard the user logs, as we get a reasonable set of covariates from the customer and transaction information.

The data sets are summarized in Table 5. As some of the customers have churned multiple times, and are therefore included multiple times, the table also includes the number of unique customers in the respective data sets.

We have a total of 15 covariates, where 7 are numeric and 8 are categorical. The numerical covariates give the time between subscriptions (for customers that have previously churned), number of previous churns, time since first registration, number of days until the current subscription expires, listed price of the current subscription, the amount paid for the subscription, and age. All numerical covariates, except the number of previous churns, were log-transformed. The categorical covariates are gender (3 categories), city (22 categories), how the customer registered (6 categories), and 5 indicator variables specifying if an individual has previously churned, if the subscription is canceled, if the age is unrealistic, if the subscription is automatically renewed, and if we do not know when the customer first registered.

Appendix C. Additional Simulations

In the following, we provide some additional simulations used to further investigate the proposed methods, and in Appendix C.3 we explain how the simulated data were generated. We again stress that the aim of the simulations is only to verify the expected behavior of our methods, and should not be interpreted as a general evaluation of them.

C.1 Non-Linear Hazards

Continuing from the simulations in Section 5, we do a simple study of the increased flexibility provided by replacing the linear predictor in a Cox regression by a neural network. To evaluate this, we extend the simulations in Section 5 by replacing $g(\mathbf{x}) = \boldsymbol{\beta}^T\mathbf{x}$ with the non-linear function

$$g(\mathbf{x}) = \boldsymbol{\beta}^T\mathbf{x} + \frac{2}{3}(x_1^2 + x_3^2 + x_1x_2 + x_1x_3 + x_2x_3), \tag{C.1}$$

where x_i denotes element i of $\mathbf{x}$. The simulations are otherwise unchanged from the linear case in Section 5. We sample 10,000 training samples, 10,000 test samples, and 1,000 samples used for validation (validation data is used for early stopping), and fit the Cox-SGD and

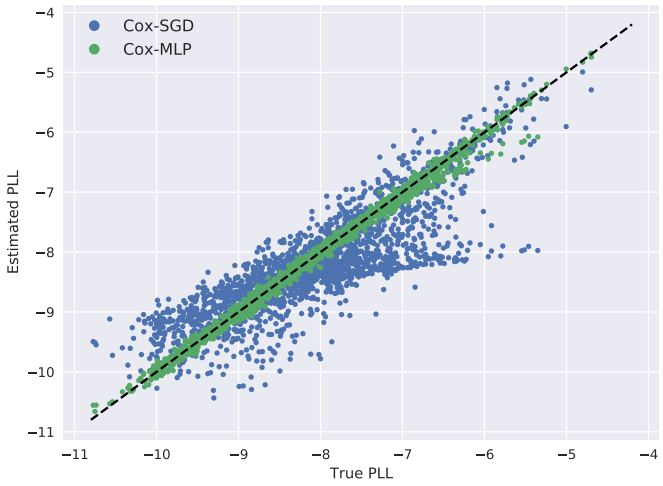


Figure C.1: Partial log-likelihood estimates of Cox-SGD and Cox-MLP plotted against the true partial log-likelihoods, for 2,000 samples of the test set.

classical Cox regression from Section 5. Additionally, we fit the Cox model in Section 3.3, where $g(\mathbf{x})$ is parameterized by a one-hidden layer MLP (multilayer perceptron) with 64 hidden nodes and ReLU activations. This model will be referred to as *Cox-MLP* (we drop (CC) as we do not consider DeepSurv here).

In Figure C.1 we have, for 2,000 individuals of the test set, plotted the individual partial log-likelihood estimates of Cox-SGD and Cox-MLP against the true individual partial log-likelihoods. That is, for each i with $D_i = 1$, we plot

$$\hat{\ell}_i = -\log\left(\sum_{j \in \mathcal{R}_i} \exp[\hat{g}(\mathbf{x}_j) - \hat{g}(\mathbf{x}_i)]\right)$$

against the true ℓ_i. The closer a method estimates $g(\mathbf{x})$ to the true predictor in (C.1), the closer the scatter plot should be to the identity function (straight line with slope 1 through the origin). As expected, we see that Cox-MLP produces likelihood estimates very close to the true likelihoods, while Cox-SGD struggles to represent this non-linear function.

C.2 Non-Proportional Hazards

In our final simulations, we investigate the effect of removing the proportionality constraint in the Cox models. Building on the previous simulations, we add a time dependent term to the risk function. The hazard function is now given by $h(t \,|\, \mathbf{x}) = h_0 \exp[g(t, \mathbf{x})]$, with $h_0 = 0.02$ and

$$\begin{aligned} g(t, \mathbf{x}) &= a(\mathbf{x}) + b(\mathbf{x})\, t, \\ a(\mathbf{x}) &= g_{ph}(\mathbf{x}) + \text{sign}(x_3), \\ b(\mathbf{x}) &= |0.2\,(x_0 + x_1) + 0.5\, x_0 x_1|, \end{aligned}$$